

Deep Learning Fundamentals

AI Learning Hub - Comprehensive Course Guide

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Neural Network Architecture

A neural network consists of layers of interconnected nodes (neurons). Each connection has a weight that is adjusted during training.

The basic architecture includes:

- Input Layer: Receives the raw data features.
- Hidden Layers: Process information through weighted connections and activation functions. Deep networks have multiple hidden layers.
- Output Layer: Produces the final prediction.

Key concepts:

- Activation Functions: ReLU ($\max(0, x)$), Sigmoid ($1/(1+e^{-x})$), Tanh, Leaky ReLU, Softmax (for multi-class classification).
- Weights and Biases: Parameters learned during training.
- Forward Propagation: Data flows from input to output through the network.

Backpropagation and Optimization

Backpropagation is the algorithm used to train neural networks by computing gradients of the loss function with respect to each weight using the chain rule.

The process:

1. Forward pass: compute predictions.
2. Compute loss using a loss function (MSE, Cross-Entropy).
3. Backward pass: compute gradients via chain rule.
4. Update weights: $w = w - \text{learning_rate} * \text{gradient}$.

Optimizers:

- SGD (Stochastic Gradient Descent): Updates weights after each sample.
- Adam (Adaptive Moment Estimation): Combines momentum and RMSprop. Uses running averages of gradients and their squares.
- Learning Rate Scheduling: Reduce LR on plateau, cosine annealing, warm restarts.

CNNs and RNNs

Convolutional Neural Networks (CNNs) are designed for spatial data like images.

- Convolutional layers apply learnable filters to detect features (edges, textures).
- Pooling layers reduce spatial dimensions (MaxPooling, AveragePooling).
- Architectures: LeNet, AlexNet, VGG, ResNet, Inception.

Recurrent Neural Networks (RNNs) handle sequential data.

- Standard RNNs suffer from vanishing gradients.
- LSTM (Long Short-Term Memory) uses gates (forget, input, output) to control information flow and capture long-range dependencies.
- GRU (Gated Recurrent Unit) is a simplified version of LSTM with two gates.